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Data Set Name	WORK.FOURTH_DOWN_CLEAN	Observations	2633
Member Type	DATA	Variables	372
Engine	V9	Indexes	0
Created	05/04/2026 21:39:12	Observation Length	4120
Last Modified	05/04/2026 21:39:12	Deleted Observations	0
Protection		Compressed	NO
Data Set Type		Sorted	NO
Label			
Data Representation	SOLARIS_X86_64, LINUX_X86_64, ALPHA_TRU64, LINUX_IA64		
Encoding	utf-8 Unicode (UTF-8)		

Engine/Host Dependent Information	
Data Set Page Size	131072
Number of Data Set Pages	86
First Data Page	1
Max Obs per Page	31
Obs in First Data Page	18
Number of Data Set Repairs	0
Filename	/saswork/SAS_work09EA00001A1B_odaws01-usw2.oda.sas.com/SAS_work656B00001A1B_odaws01-usw2.oda.sas.com/fourth_down_clean.sas7bdat
Release Created	9.0401M8
Host Created	Linux
Inode Number	1074798019
Access Permission	rw-r--r--
Owner Name	u64427608
File Size	11MB
File Size (bytes)	11403264

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Alphabetic List of Variables and Attributes					
#	Variable	Type	Len	Format	Informat
199	VAR199	Char	1	\$1.	\$1.
340	aborted_play	Num	8	BEST12.	BEST32.
81	air_epa	Num	8	BEST12.	BEST32.
108	air_wpa	Num	8	BEST12.	BEST32.
39	air_yards	Num	8	BEST12.	BEST32.
166	assist_tackle	Num	8	BEST12.	BEST32.
228	assist_tackle_1_player_id	Char	10	\$10.	\$10.
229	assist_tackle_1_player_name	Char	19	\$19.	\$19.
230	assist_tackle_1_team	Char	3	\$3.	\$3.
231	assist_tackle_2_player_id	Char	10	\$10.	\$10.
232	assist_tackle_2_player_name	Char	19	\$19.	\$19.
233	assist_tackle_2_team	Char	3	\$3.	\$3.
234	assist_tackle_3_player_id	Char	10	\$10.	\$10.
235	assist_tackle_3_player_name	Char	12	\$12.	\$12.
236	assist_tackle_3_team	Char	3	\$3.	\$3.
237	assist_tackle_4_player_id	Char	10	\$10.	\$10.
238	assist_tackle_4_player_name	Char	9	\$9.	\$9.
239	assist_tackle_4_team	Char	3	\$3.	\$3.
337	away_coach	Char	20	\$20.	\$20.
324	away_score	Num	8	BEST12.	BEST32.
5	away_team	Char	3	\$3.	\$3.
48	away_timeouts_remaining	Num	8	BEST12.	BEST32.
96	away_wp	Num	8	BEST12.	BEST32.
101	away_wp_post	Num	8	BEST12.	BEST32.
206	blocked_player_id	Char	10	\$10.	\$10.
207	blocked_player_name	Char	11	\$11.	\$11.
83	comp_air_epa	Num	8	BEST12.	BEST32.
110	comp_air_wpa	Num	8	BEST12.	BEST32.
84	comp_yac_epa	Num	8	BEST12.	BEST32.
111	comp_yac_wpa	Num	8	BEST12.	BEST32.
165	complete_pass	Num	8	BEST12.	BEST32.
287	cp	Num	8	BEST12.	BEST32.
288	cpoe	Num	8	BEST12.	BEST32.
94	def_wp	Num	8	BEST12.	BEST32.
282	defensive_extra_point_attempt	Num	8	BEST12.	BEST32.
283	defensive_extra_point_conv	Num	8	BEST12.	BEST32.

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Alphabetic List of Variables and Attributes					
#	Variable	Type	Len	Format	Informat
280	defensive_two_point_attempt	Num	8	BEST12.	BEST32.
281	defensive_two_point_conv	Num	8	BEST12.	BEST32.
10	defteam	Char	3	\$3.	\$3.
59	defteam_score	Num	8	BEST12.	BEST32.
62	defteam_score_post	Num	8	BEST12.	BEST32.
55	defteam_timeouts_remaining	Num	8	BEST12.	BEST32.
28	desc	Char	719	\$719.	\$719.
331	div_game	Num	8	BEST12.	BEST32.
22	down	Num	8	BEST12.	BEST32.
19	drive	Num	8	BEST12.	BEST32.
317	drive_end_transition	Char	21	\$21.	\$21.
321	drive_end_yard_line	Char	6	\$6.	\$6.
312	drive_ended_with_score	Num	8	BEST12.	BEST32.
310	drive_first_downs	Num	8	BEST12.	BEST32.
319	drive_game_clock_end	Num	8	TIME20.3	TIME20.3
318	drive_game_clock_start	Num	8	TIME20.3	TIME20.3
311	drive_inside20	Num	8	BEST12.	BEST32.
308	drive_play_count	Num	8	BEST12.	BEST32.
323	drive_play_id_ended	Num	8	BEST12.	BEST32.
322	drive_play_id_started	Num	8	BEST12.	BEST32.
314	drive_quarter_end	Num	8	BEST12.	BEST32.
313	drive_quarter_start	Num	8	BEST12.	BEST32.
307	drive_real_start_time	Num	8	B8601DZ35.	B8601DZ35.
316	drive_start_transition	Char	21	\$21.	\$21.
320	drive_start_yard_line	Char	6	\$6.	\$6.
309	drive_time_of_possession	Num	8	TIME20.3	TIME20.3
315	drive_yards_penalized	Num	8	BEST12.	BEST32.
303	end_clock_time	Num	8	B8601DZ35.	B8601DZ35.
304	end_yard_line	Char	1	\$1.	\$1.
73	ep	Num	8	BEST12.	BEST32.
74	epa	Num	8	BEST12.	BEST32.
159	extra_point_attempt	Num	8	BEST12.	BEST32.
71	extra_point_prob	Num	8	BEST12.	BEST32.
45	extra_point_result	Char	7	\$7.	\$7.
361	fantasy	Char	18	\$18.	\$18.
362	fantasy_id	Char	10	\$10.	\$10.

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Alphabetic List of Variables and Attributes					
#	Variable	Type	Len	Format	Informat
360	fantasy_player_id	Char	10	\$10.	\$10.
359	fantasy_player_name	Char	18	\$18.	\$18.
68	fg_prob	Num	8	BEST12.	BEST32.
161	field_goal_attempt	Num	8	BEST12.	BEST32.
43	field_goal_result	Char	7	\$7.	\$7.
350	first_down	Num	8	BEST12.	BEST32.
122	first_down_pass	Num	8	BEST12.	BEST32.
123	first_down_penalty	Num	8	BEST12.	BEST32.
121	first_down_rush	Num	8	BEST12.	BEST32.
305	fixed_drive	Num	8	BEST12.	BEST32.
306	fixed_drive_result	Char	17	\$17.	\$17.
217	forced_fumble_player_1_player_id	Char	10	\$10.	\$10.
218	forced_fumble_player_1_player_na	Char	17	\$17.	\$17.
216	forced_fumble_player_1_team	Char	3	\$3.	\$3.
220	forced_fumble_player_2_player_id	Char	10	\$10.	\$10.
221	forced_fumble_player_2_player_na	Char	11	\$11.	\$11.
219	forced_fumble_player_2_team	Char	3	\$3.	\$3.
126	fourth_down_converted	Num	8	BEST12.	BEST32.
127	fourth_down_failed	Num	8	BEST12.	BEST32.
164	fumble	Num	8	BEST12.	BEST32.
141	fumble_forced	Num	8	BEST12.	BEST32.
148	fumble_lost	Num	8	BEST12.	BEST32.
142	fumble_not_forced	Num	8	BEST12.	BEST32.
143	fumble_out_of_bounds	Num	8	BEST12.	BEST32.
259	fumble_recovery_1_player_id	Char	10	\$10.	\$10.
260	fumble_recovery_1_player_name	Char	18	\$18.	\$18.
257	fumble_recovery_1_team	Char	3	\$3.	\$3.
258	fumble_recovery_1_yards	Num	8	BEST12.	BEST32.
263	fumble_recovery_2_player_id	Char	10	\$10.	\$10.
264	fumble_recovery_2_player_name	Char	10	\$10.	\$10.
261	fumble_recovery_2_team	Char	3	\$3.	\$3.
262	fumble_recovery_2_yards	Num	8	BEST12.	BEST32.
252	fumbled_1_player_id	Char	10	\$10.	\$10.
253	fumbled_1_player_name	Char	18	\$18.	\$18.
251	fumbled_1_team	Char	3	\$3.	\$3.
254	fumbled_2_player_id	Char	10	\$10.	\$10.

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Alphabetic List of Variables and Attributes					
#	Variable	Type	Len	Format	Informat
255	fumbled_2_player_name	Char	10	\$10.	\$10.
256	fumbled_2_team	Char	3	\$3.	\$3.
13	game_date	Num	8	YYMMDD10.	YYMMDD10.
17	game_half	Char	8	\$8.	\$8.
2	game_id	Char	15	\$15.	\$15.
16	game_seconds_remaining	Num	8	BEST12.	BEST32.
339	game_stadium	Char	31	\$31.	\$31.
23	goal_to_go	Num	8	BEST12.	BEST32.
267	half_sack_1_player_id	Char	10	\$10.	\$10.
268	half_sack_1_player_name	Char	16	\$16.	\$16.
269	half_sack_2_player_id	Char	10	\$10.	\$10.
270	half_sack_2_player_name	Char	13	\$13.	\$13.
15	half_seconds_remaining	Num	8	BEST12.	BEST32.
336	home_coach	Char	20	\$20.	\$20.
364	home_opening_kickoff	Num	8	BEST12.	BEST32.
325	home_score	Num	8	BEST12.	BEST32.
4	home_team	Char	3	\$3.	\$3.
47	home_timeouts_remaining	Num	8	BEST12.	BEST32.
95	home_wp	Num	8	BEST12.	BEST32.
100	home_wp_post	Num	8	BEST12.	BEST32.
358	id	Char	10	\$10.	\$10.
128	incomplete_pass	Num	8	BEST12.	BEST32.
130	interception	Num	8	BEST12.	BEST32.
188	interception_player_id	Char	10	\$10.	\$10.
189	interception_player_name	Char	17	\$17.	\$17.
357	jersey_number	Num	8	BEST12.	BEST32.
44	kick_distance	Num	8	BEST12.	BEST32.
203	kicker_player_id	Char	10	\$10.	\$10.
202	kicker_player_name	Char	12	\$12.	\$12.
162	kickoff_attempt	Num	8	BEST12.	BEST32.
139	kickoff_downed	Num	8	BEST12.	BEST32.
140	kickoff_fair_catch	Num	8	BEST12.	BEST32.
137	kickoff_in_endzone	Num	8	BEST12.	BEST32.
136	kickoff_inside_twenty	Num	8	BEST12.	BEST32.
138	kickoff_out_of_bounds	Num	8	BEST12.	BEST32.
197	kickoff_returner_player_id	Char	10	\$10.	\$10.

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Alphabetic List of Variables and Attributes					
#	Variable	Type	Len	Format	Informat
196	kickoff_returner_player_name	Char	16	\$16.	\$16.
190	lateral_interception_player_id	Char	10	\$10.	\$10.
191	lateral_interception_player_name	Char	8	\$8.	\$8.
198	lateral_kickoff_returner_player_	Char	1	\$1.	\$1.
194	lateral_punt_returner_player_id	Char	1	\$1.	\$1.
195	lateral_punt_returner_player_nam	Char	1	\$1.	\$1.
180	lateral_receiver_player_id	Char	10	\$10.	\$10.
181	lateral_receiver_player_name	Char	11	\$11.	\$11.
182	lateral_receiving_yards	Num	8	BEST12.	BEST32.
167	lateral_reception	Num	8	BEST12.	BEST32.
170	lateral_recovery	Num	8	BEST12.	BEST32.
169	lateral_return	Num	8	BEST12.	BEST32.
168	lateral_rush	Num	8	BEST12.	BEST32.
183	lateral_rusher_player_id	Char	10	\$10.	\$10.
184	lateral_rusher_player_name	Char	7	\$7.	\$7.
185	lateral_rushing_yards	Num	8	BEST12.	BEST32.
186	lateral_sack_player_id	Char	1	\$1.	\$1.
187	lateral_sack_player_name	Char	1	\$1.	\$1.
326	location	Char	7	\$7.	\$7.
356	name	Char	17	\$17.	\$17.
297	nfl_api_id	Char	36	\$36.	\$36.
32	no_huddle	Num	8	BEST12.	BEST32.
64	no_score_prob	Num	8	BEST12.	BEST32.
3	old_game_id	Num	8	BEST12.	BEST32.
65	opp_fg_prob	Num	8	BEST12.	BEST32.
66	opp_safety_prob	Num	8	BEST12.	BEST32.
67	opp_td_prob	Num	8	BEST12.	BEST32.
292	order_sequence	Num	8	BEST12.	BEST32.
363	out_of_bounds	Num	8	BEST12.	BEST32.
149	own_kickoff_recovery	Num	8	BEST12.	BEST32.
204	own_kickoff_recovery_player_id	Char	10	\$10.	\$10.
205	own_kickoff_recovery_player_name	Char	12	\$12.	\$12.
150	own_kickoff_recovery_td	Num	8	BEST12.	BEST32.
348	pass	Num	8	BEST12.	BEST32.
153	pass_attempt	Num	8	BEST12.	BEST32.
247	pass_defense_1_player_id	Char	10	\$10.	\$10.

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Alphabetic List of Variables and Attributes					
#	Variable	Type	Len	Format	Informat
248	pass_defense_1_player_name	Char	18	\$18.	\$18.
249	pass_defense_2_player_id	Char	10	\$10.	\$10.
250	pass_defense_2_player_name	Char	17	\$17.	\$17.
37	pass_length	Char	5	\$5.	\$5.
38	pass_location	Char	6	\$6.	\$6.
372	pass_oe	Num	8	BEST12.	BEST32.
156	pass_touchdown	Num	8	BEST12.	BEST32.
342	passer	Char	13	\$13.	\$13.
353	passer_id	Char	10	\$10.	\$10.
343	passer_jersey_number	Num	8	BEST12.	BEST32.
171	passer_player_id	Char	10	\$10.	\$10.
172	passer_player_name	Char	13	\$13.	\$13.
173	passing_yards	Num	8	BEST12.	BEST32.
146	penalty	Num	8	BEST12.	BEST32.
274	penalty_player_id	Char	10	\$10.	\$10.
275	penalty_player_name	Char	18	\$18.	\$18.
273	penalty_team	Char	3	\$3.	\$3.
279	penalty_type	Char	31	\$31.	\$31.
276	penalty_yards	Num	8	BEST12.	BEST32.
352	play	Num	8	BEST12.	BEST32.
298	play_clock	Num	8	BEST12.	BEST32.
299	play_deleted	Num	8	BEST12.	BEST32.
1	play_id	Num	8	BEST12.	BEST32.
29	play_type	Char	11	\$11.	\$11.
300	play_type_nfl	Char	28	\$28.	\$28.
8	posteam	Char	3	\$3.	\$3.
58	posteam_score	Num	8	BEST12.	BEST32.
61	posteam_score_post	Num	8	BEST12.	BEST32.
54	posteam_timeouts_remaining	Num	8	BEST12.	BEST32.
9	posteam_type	Char	4	\$4.	\$4.
163	punt_attempt	Num	8	BEST12.	BEST32.
120	punt_blocked	Num	8	BEST12.	BEST32.
134	punt_downed	Num	8	BEST12.	BEST32.
135	punt_fair_catch	Num	8	BEST12.	BEST32.
132	punt_in_endzone	Num	8	BEST12.	BEST32.
131	punt_inside_twenty	Num	8	BEST12.	BEST32.

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Alphabetic List of Variables and Attributes					
#	Variable	Type	Len	Format	Informat
133	punt_out_of_bounds	Num	8	BEST12.	BEST32.
192	punt_returner_player_id	Char	10	\$10.	\$10.
193	punt_returner_player_name	Char	16	\$16.	\$16.
200	punter_player_id	Char	10	\$10.	\$10.
201	punter_player_name	Char	12	\$12.	\$12.
33	qb_dropback	Num	8	BEST12.	BEST32.
365	qb_epa	Num	8	BEST12.	BEST32.
151	qb_hit	Num	8	BEST12.	BEST32.
212	qb_hit_1_player_id	Char	10	\$10.	\$10.
213	qb_hit_1_player_name	Char	18	\$18.	\$18.
214	qb_hit_2_player_id	Char	10	\$10.	\$10.
215	qb_hit_2_player_name	Char	13	\$13.	\$13.
34	qb_kneel	Num	8	BEST12.	BEST32.
36	qb_scrumble	Num	8	BEST12.	BEST32.
35	qb_spike	Num	8	BEST12.	BEST32.
21	qtr	Num	8	BEST12.	BEST32.
18	quarter_end	Num	8	BEST12.	BEST32.
14	quarter_seconds_remaining	Num	8	BEST12.	BEST32.
346	receiver	Char	18	\$18.	\$18.
355	receiver_id	Char	10	\$10.	\$10.
347	receiver_jersey_number	Num	8	BEST12.	BEST32.
174	receiver_player_id	Char	10	\$10.	\$10.
175	receiver_player_name	Char	18	\$18.	\$18.
176	receiving_yards	Num	8	BEST12.	BEST32.
277	replay_or_challenge	Num	8	BEST12.	BEST32.
278	replay_or_challenge_result	Char	8	\$8.	\$8.
327	result	Num	8	BEST12.	BEST32.
271	return_team	Char	3	\$3.	\$3.
158	return_touchdown	Num	8	BEST12.	BEST32.
272	return_yards	Num	8	BEST12.	BEST32.
332	roof	Char	8	\$8.	\$8.
42	run_gap	Char	6	\$6.	\$6.
41	run_location	Char	6	\$6.	\$6.
349	rush	Num	8	BEST12.	BEST32.
152	rush_attempt	Num	8	BEST12.	BEST32.
157	rush_touchdown	Num	8	BEST12.	BEST32.

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Alphabetic List of Variables and Attributes					
#	Variable	Type	Len	Format	Informat
344	rusher	Char	17	\$17.	\$17.
354	rusher_id	Char	10	\$10.	\$10.
345	rusher_jersey_number	Num	8	BEST12.	BEST32.
177	rusher_player_id	Char	10	\$10.	\$10.
178	rusher_player_name	Char	17	\$17.	\$17.
179	rushing_yards	Num	8	BEST12.	BEST32.
154	sack	Num	8	BEST12.	BEST32.
265	sack_player_id	Char	10	\$10.	\$10.
266	sack_player_name	Char	18	\$18.	\$18.
145	safety	Num	8	BEST12.	BEST32.
285	safety_player_id	Char	10	\$10.	\$10.
284	safety_player_name	Char	14	\$14.	\$14.
69	safety_prob	Num	8	BEST12.	BEST32.
60	score_differential	Num	8	BEST12.	BEST32.
63	score_differential_post	Num	8	BEST12.	BEST32.
286	season	Num	8	BEST12.	BEST32.
6	season_type	Char	4	\$4.	\$4.
289	series	Num	8	BEST12.	BEST32.
291	series_result	Char	17	\$17.	\$17.
290	series_success	Num	8	BEST12.	BEST32.
31	shotgun	Num	8	BEST12.	BEST32.
11	side_of_field	Char	3	\$3.	\$3.
144	solo_tackle	Num	8	BEST12.	BEST32.
224	solo_tackle_1_player_id	Char	10	\$10.	\$10.
226	solo_tackle_1_player_name	Char	19	\$19.	\$19.
222	solo_tackle_1_team	Char	3	\$3.	\$3.
225	solo_tackle_2_player_id	Char	10	\$10.	\$10.
227	solo_tackle_2_player_name	Char	13	\$13.	\$13.
223	solo_tackle_2_team	Char	3	\$3.	\$3.
20	sp	Num	8	BEST12.	BEST32.
351	special	Num	8	BEST12.	BEST32.
301	special_teams_play	Num	8	BEST12.	BEST32.
329	spread_line	Num	8	BEST12.	BEST32.
302	st_play_type	Char	1	\$1.	\$1.
295	stadium	Char	31	\$31.	\$31.
338	stadium_id	Char	5	\$5.	\$5.

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Alphabetic List of Variables and Attributes					
#	Variable	Type	Len	Format	Informat
293	start_time	Char	20	\$20.	\$20.
341	success	Num	8	BEST12.	BEST32.
333	surface	Char	10	\$10.	\$10.
208	tackle_for_loss_1_player_id	Char	10	\$10.	\$10.
209	tackle_for_loss_1_player_name	Char	18	\$18.	\$18.
210	tackle_for_loss_2_player_id	Char	1	\$1.	\$1.
211	tackle_for_loss_2_player_name	Char	1	\$1.	\$1.
240	tackle_with_assist	Num	8	BEST12.	BEST32.
241	tackle_with_assist_1_player_id	Char	10	\$10.	\$10.
242	tackle_with_assist_1_player_name	Char	18	\$18.	\$18.
243	tackle_with_assist_1_team	Char	3	\$3.	\$3.
244	tackle_with_assist_2_player_id	Char	1	\$1.	\$1.
245	tackle_with_assist_2_player_name	Char	1	\$1.	\$1.
246	tackle_with_assist_2_team	Char	1	\$1.	\$1.
147	tackled_for_loss	Num	8	BEST12.	BEST32.
53	td_player_id	Char	10	\$10.	\$10.
52	td_player_name	Char	18	\$18.	\$18.
70	td_prob	Num	8	BEST12.	BEST32.
51	td_team	Char	3	\$3.	\$3.
334	temp	Num	8	BEST12.	BEST32.
124	third_down_converted	Num	8	BEST12.	BEST32.
125	third_down_failed	Num	8	BEST12.	BEST32.
24	time	Num	8	TIME20.3	TIME20.3
294	time_of_day	Num	8	B8601DZ35.	B8601DZ35.
49	timeout	Num	8	BEST12.	BEST32.
50	timeout_team	Char	3	\$3.	\$3.
328	total	Num	8	BEST12.	BEST32.
86	total_away_comp_air_epa	Num	8	BEST12.	BEST32.
113	total_away_comp_air_wpa	Num	8	BEST12.	BEST32.
88	total_away_comp_yac_epa	Num	8	BEST12.	BEST32.
115	total_away_comp_yac_wpa	Num	8	BEST12.	BEST32.
76	total_away_epa	Num	8	BEST12.	BEST32.
80	total_away_pass_epa	Num	8	BEST12.	BEST32.
107	total_away_pass_wpa	Num	8	BEST12.	BEST32.
90	total_away_raw_air_epa	Num	8	BEST12.	BEST32.
117	total_away_raw_air_wpa	Num	8	BEST12.	BEST32.

The CONTENTS Procedure

Alphabetic List of Variables and Attributes					
#	Variable	Type	Len	Format	Informat
92	total_away_raw_yac_epa	Num	8	BEST12.	BEST32.
119	total_away_raw_yac_wpa	Num	8	BEST12.	BEST32.
78	total_away_rush_epa	Num	8	BEST12.	BEST32.
105	total_away_rush_wpa	Num	8	BEST12.	BEST32.
57	total_away_score	Num	8	BEST12.	BEST32.
85	total_home_comp_air_epa	Num	8	BEST12.	BEST32.
112	total_home_comp_air_wpa	Num	8	BEST12.	BEST32.
87	total_home_comp_yac_epa	Num	8	BEST12.	BEST32.
114	total_home_comp_yac_wpa	Num	8	BEST12.	BEST32.
75	total_home_epa	Num	8	BEST12.	BEST32.
79	total_home_pass_epa	Num	8	BEST12.	BEST32.
106	total_home_pass_wpa	Num	8	BEST12.	BEST32.
89	total_home_raw_air_epa	Num	8	BEST12.	BEST32.
116	total_home_raw_air_wpa	Num	8	BEST12.	BEST32.
91	total_home_raw_yac_epa	Num	8	BEST12.	BEST32.
118	total_home_raw_yac_wpa	Num	8	BEST12.	BEST32.
77	total_home_rush_epa	Num	8	BEST12.	BEST32.
104	total_home_rush_wpa	Num	8	BEST12.	BEST32.
56	total_home_score	Num	8	BEST12.	BEST32.
330	total_line	Num	8	BEST12.	BEST32.
129	touchback	Num	8	BEST12.	BEST32.
155	touchdown	Num	8	BEST12.	BEST32.
160	two_point_attempt	Num	8	BEST12.	BEST32.
46	two_point_conv_result	Char	7	\$7.	\$7.
72	two_point_conversion_prob	Num	8	BEST12.	BEST32.
103	vegas_home_wp	Num	8	BEST12.	BEST32.
99	vegas_home_wpa	Num	8	BEST12.	BEST32.
102	vegas_wp	Num	8	BEST12.	BEST32.
98	vegas_wpa	Num	8	BEST12.	BEST32.
296	weather	Char	75	\$75.	\$75.
7	week	Num	8	BEST12.	BEST32.
335	wind	Num	8	BEST12.	BEST32.
93	wp	Num	8	BEST12.	BEST32.
97	wpa	Num	8	BEST12.	BEST32.
371	xpass	Num	8	BEST12.	BEST32.
366	xyac_epa	Num	8	BEST12.	BEST32.

The CONTENTS Procedure

Alphabetic List of Variables and Attributes					
#	Variable	Type	Len	Format	Informat
370	xyac_fd	Num	8	BEST12.	BEST32.
367	xyac_mean_yardage	Num	8	BEST12.	BEST32.
368	xyac_median_yardage	Num	8	BEST12.	BEST32.
369	xyac_success	Num	8	BEST12.	BEST32.
82	yac_epa	Num	8	BEST12.	BEST32.
109	yac_wpa	Num	8	BEST12.	BEST32.
12	yardline_100	Num	8	BEST12.	BEST32.
40	yards_after_catch	Num	8	BEST12.	BEST32.
30	yards_gained	Num	8	BEST12.	BEST32.
27	ydsnet	Num	8	BEST12.	BEST32.
26	ydstogo	Num	8	BEST12.	BEST32.
25	yrdln	Char	6	\$6.	\$6.

Obs	posteam	down	play_type	qtr	ydstogo	yardline_100	score_differential
1	BAL	4	field_goal	1	22	34	-7
2	BUF	4	punt	1	3	58	4
3	BUF	4	punt	2	6	73	-3
4	BUF	4	field_goal	2	5	7	-10
5	BAL	4	field_goal	2	24	31	7
6	BUF	4	no_play	3	5	31	-14
7	BAL	4	punt	3	10	67	8
8	BUF	4	punt	3	8	71	-8
9	BUF	4	punt	4	16	63	-15
10	BAL	4	punt	4	4	59	15

The FREQ Procedure

posteam	Frequency	Percent	Cumulative Frequency	Cumulative Percent
ATL	137	5.20	137	5.20
BAL	118	4.48	255	9.68
BUF	122	4.63	377	14.32
CIN	124	4.71	501	19.03
CLE	157	5.96	658	24.99
DAL	123	4.67	781	29.66
DEN	150	5.70	931	35.36
GB	114	4.33	1045	39.69
IND	114	4.33	1159	44.02
KC	125	4.75	1284	48.77
LA	141	5.36	1425	54.12
MIA	119	4.52	1544	58.64
MIN	128	4.86	1672	63.50
NE	148	5.62	1820	69.12
NYJ	145	5.51	1965	74.63
PHI	150	5.70	2115	80.33
PIT	135	5.13	2250	85.45
SEA	137	5.20	2387	90.66
SF	111	4.22	2498	94.87
TB	135	5.13	2633	100.00

The FREQ Procedure

go_for_it	Frequency	Percent	Cumulative Frequency	Cumulative Percent
Kick	1905	77.63	1905	77.63
Go For It	549	22.37	2454	100.00

The MEANS Procedure

Variable	N	N Miss	Mean	Std Dev	Minimum	Maximum
ydstogo	2454	0	7.4779951	5.6181782	1.0000000	32.0000000
yardline_100	2454	0	47.4682152	24.5344008	1.0000000	99.0000000
score_differential	2454	0	-0.2950285	10.2403724	-39.0000000	39.0000000
game_seconds_remaining	2454	0	1656.75	1042.74	1.0000000	3557.00
half_seconds_remaining	2454	0	772.8859006	530.1998770	1.0000000	1785.00
posteam_timeouts_remaining	2454	0	2.5419723	0.7817676	0	3.0000000
defteam_timeouts_remaining	2454	0	2.5256724	0.8304937	0	3.0000000

The FREQ Procedure

qtr	Frequency	Percent	Cumulative Frequency	Cumulative Percent
1	501	20.42	501	20.42
2	704	28.69	1205	49.10
3	512	20.86	1717	69.97
4	723	29.46	2440	99.43
5	14	0.57	2454	100.00

The FREQ Procedure

Frequency Percent Row Pct Col Pct	Table of go_for_it by qtr						
	go_for_it	qtr					Total
		1	2	3	4	5	
Kick		406	588	419	481	11	1905
		16.54	23.96	17.07	19.60	0.45	77.63
		21.31	30.87	21.99	25.25	0.58	
		81.04	83.52	81.84	66.53	78.57	
Go For It		95	116	93	242	3	549
		3.87	4.73	3.79	9.86	0.12	22.37
		17.30	21.13	16.94	44.08	0.55	
		18.96	16.48	18.16	33.47	21.43	
Total		501	704	512	723	14	2454
		20.42	28.69	20.86	29.46	0.57	100.00

Statistics for Table of go_for_it by qtr

Statistic	DF	Value	Prob
Chi-Square	4	73.9582	<.0001
Likelihood Ratio Chi-Square	4	70.6564	<.0001
Mantel-Haenszel Chi-Square	1	44.7601	<.0001
Phi Coefficient		0.1736	
Contingency Coefficient		0.1710	
Cramer's V		0.1736	

Statistic	Value	ASE
Gamma	0.2433	0.0366
Kendall's Tau-b	0.1244	0.0189
Stuart's Tau-c	0.1266	0.0195
Somers' D C R	0.1823	0.0276
Somers' D R C	0.0849	0.0131
Pearson Correlation	0.1351	0.0206
Spearman Correlation	0.1362	0.0207
Lambda Asymmetric C R	0.0618	0.0183
Lambda Asymmetric R C	0.0000	0.0000
Lambda Symmetric	0.0469	0.0140
Uncertainty Coefficient C R	0.0103	0.0025
Uncertainty Coefficient R C	0.0271	0.0065
Uncertainty Coefficient Symmetric	0.0149	0.0036

Sample Size = 2454

The FREQ Procedure

Frequency Percent Row Pct Col Pct	Table of go_for_it by ydstogo_bin				
	go_for_it	ydstogo_bin			
		Long	Medium	Short	Total
	Kick	1288 52.49 67.61 92.33	441 17.97 23.15 78.05	176 7.17 9.24 35.63	1905 77.63
	Go For It	107 4.36 19.49 7.67	124 5.05 22.59 21.95	318 12.96 57.92 64.37	549 22.37
	Total	1395 56.85	565 23.02	494 20.13	2454 100.00

Statistics for Table of go_for_it by ydstogo_bin

Statistic	DF	Value	Prob
Chi-Square	2	675.4608	<.0001
Likelihood Ratio Chi-Square	2	615.7894	<.0001
Mantel-Haenszel Chi-Square	1	628.7466	<.0001
Phi Coefficient		0.5246	
Contingency Coefficient		0.4646	
Cramer's V		0.5246	

Statistic	Value	ASE
Gamma	0.7797	0.0188
Kendall's Tau-b	0.4587	0.0175
Stuart's Tau-c	0.4129	0.0187
Somers' D C R	0.5944	0.0218
Somers' D R C	0.3540	0.0153
Pearson Correlation	0.5063	0.0187
Spearman Correlation	0.4809	0.0184
Lambda Asymmetric C R	0.1992	0.0174
Lambda Asymmetric R C	0.2587	0.0349
Lambda Symmetric	0.2195	0.0219
Uncertainty Coefficient C R	0.1278	0.0099
Uncertainty Coefficient R C	0.2360	0.0177
Uncertainty Coefficient Symmetric	0.1658	0.0127

Sample Size = 2454

The FREQ Procedure

Frequency Percent Row Pct Col Pct	Table of go_for_it by score_bin				
	go_for_it	score_bin			
		Close	Leading	Trailing	Total
	Kick	1306 53.22 68.56 80.07	346 14.10 18.16 86.28	253 10.31 13.28 59.95	1905 77.63
	Go For It	325 13.24 59.20 19.93	55 2.24 10.02 13.72	169 6.89 30.78 40.05	549 22.37
	Total	1631 66.46	401 16.34	422 17.20	2454 100.00

Statistics for Table of go_for_it by score_bin

Statistic	DF	Value	Prob
Chi-Square	2	98.8345	<.0001
Likelihood Ratio Chi-Square	2	91.1748	<.0001
Mantel-Haenszel Chi-Square	1	51.7538	<.0001
Phi Coefficient		0.2007	
Contingency Coefficient		0.1968	
Cramer's V		0.2007	

Statistic	Value	ASE
Gamma	0.2545	0.0432
Kendall's Tau-b	0.1133	0.0210
Stuart's Tau-c	0.0946	0.0178
Somers' D C R	0.1362	0.0253
Somers' D R C	0.0942	0.0175
Pearson Correlation	0.1453	0.0224
Spearman Correlation	0.1177	0.0218
Lambda Asymmetric C R	0.0000	0.0000
Lambda Asymmetric R C	0.0000	0.0000
Lambda Symmetric	0.0000	0.0000
Uncertainty Coefficient C R	0.0213	0.0046
Uncertainty Coefficient R C	0.0349	0.0075
Uncertainty Coefficient Symmetric	0.0265	0.0057

Sample Size = 2454

The FREQ Procedure

Frequency
Percent
Row Pct
Col Pct

Table of go_for_it by field_bin				
go_for_it	field_bin			
	Midfield	Own Territory	Scoring Range	Total
Kick	754 30.73 39.58 75.40	591 24.08 31.02 92.34	560 22.82 29.40 68.80	1905 77.63
Go For It	246 10.02 44.81 24.60	49 2.00 8.93 7.66	254 10.35 46.27 31.20	549 22.37
Total	1000 40.75	640 26.08	814 33.17	2454 100.00

Statistics for Table of go_for_it by field_bin

Statistic	DF	Value	Prob
Chi-Square	2	119.2236	<.0001
Likelihood Ratio Chi-Square	2	136.6467	<.0001
Mantel-Haenszel Chi-Square	1	7.8708	0.0050
Phi Coefficient		0.2204	
Contingency Coefficient		0.2152	
Cramer's V		0.2204	

Statistic	Value	ASE
Gamma	0.0987	0.0431
Kendall's Tau-b	0.0473	0.0208
Stuart's Tau-c	0.0452	0.0199
Somers' D C R	0.0650	0.0286
Somers' D R C	0.0344	0.0152
Pearson Correlation	0.0566	0.0218
Spearman Correlation	0.0501	0.0220
Lambda Asymmetric C R	0.0055	0.0153
Lambda Asymmetric R C	0.0000	0.0000
Lambda Symmetric	0.0040	0.0111
Uncertainty Coefficient C R	0.0257	0.0040
Uncertainty Coefficient R C	0.0524	0.0080
Uncertainty Coefficient Symmetric	0.0345	0.0053

Sample Size = 2454

The FREQ Procedure

Frequency Percent Row Pct Col Pct	Table of short_yardage by go_for_it		
	short_yardage	go_for_it	
		Kick	Go For It
	No	1729 70.46 88.21 90.76	231 9.41 11.79 42.08
	Yes	176 7.17 35.63 9.24	318 12.96 64.37 57.92
	Total	1905 77.63	549 22.37
			2454 100.00

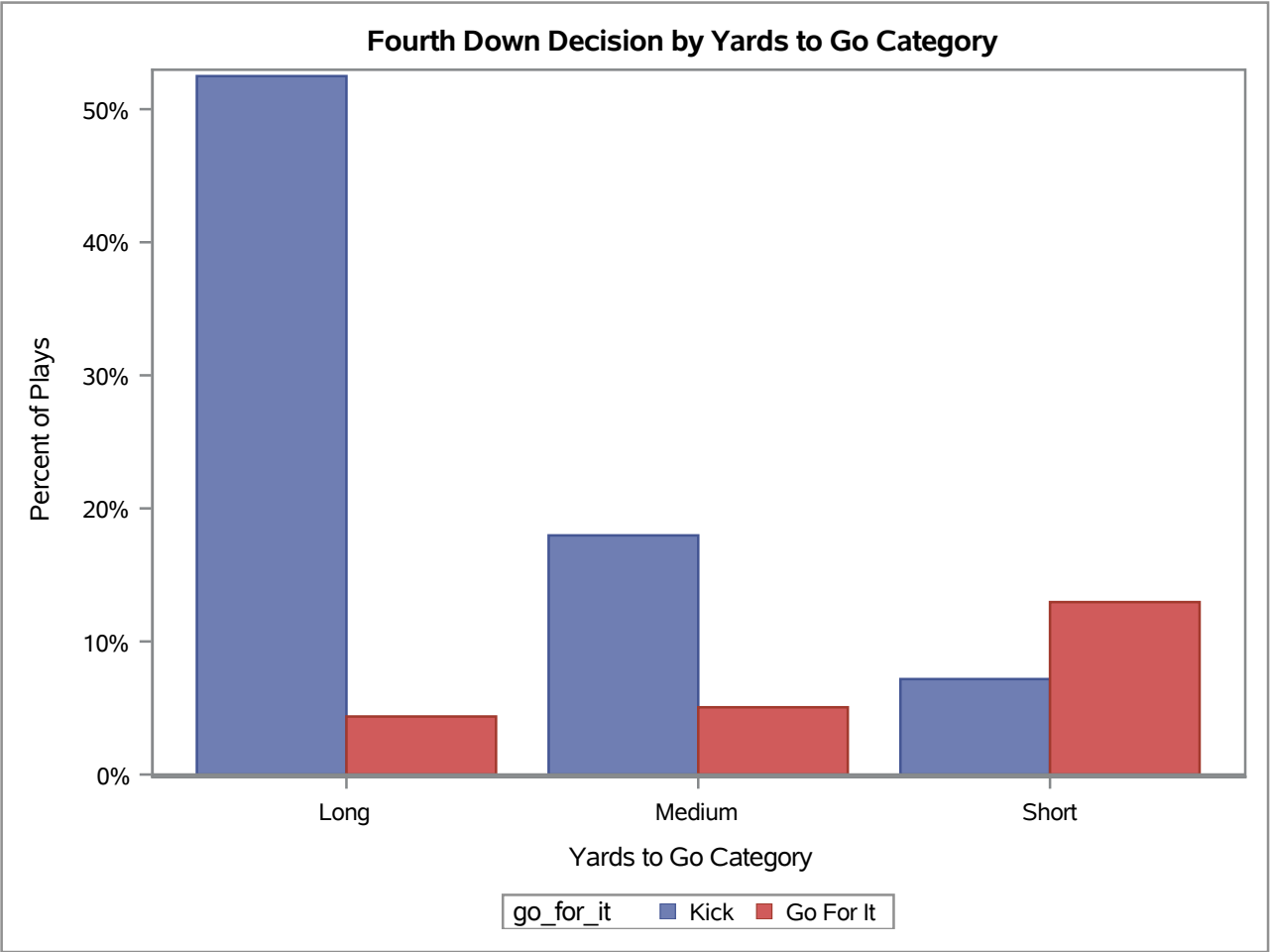
Statistics for Table of short_yardage by go_for_it

Statistic	DF	Value	Prob
Chi-Square	1	628.2652	<.0001
Likelihood Ratio Chi-Square	1	543.9994	<.0001
Continuity Adj. Chi-Square	1	625.2408	<.0001
Mantel-Haenszel Chi-Square	1	628.0092	<.0001
Phi Coefficient		0.5060	
Contingency Coefficient		0.4515	
Cramer's V		0.5060	

Fisher's Exact Test	
Cell (1,1) Frequency (F)	1729
Left-sided Pr <= F	1.0000
Right-sided Pr >= F	<.0001
Table Probability (P)	<.0001
Two-sided Pr <= P	<.0001

Odds Ratio and Relative Risks			
Statistic	Value	95% Confidence Limits	
Odds Ratio	13.5238	10.7484	17.0157
Relative Risk (Column 1)	2.4760	2.1968	2.7907
Relative Risk (Column 2)	0.1831	0.1595	0.2101

Sample Size = 2454



Initial Logistic Regression Model for Fourth Down Decisions

The LOGISTIC Procedure

Model Information	
Data Set	WORK.FOURTH_DOWN_MODEL
Response Variable	go_for_it
Number of Response Levels	2
Model	binary logit
Optimization Technique	Fisher's scoring

Number of Observations Read	2454
Number of Observations Used	2454

Response Profile		
Ordered Value	go_for_it	Total Frequency
1	0	1905
2	1	549

Probability modeled is go_for_it=1.

Class Level Information					
Class	Value	Design Variables			
qtr	1	0	0	0	0
	2	1	0	0	0
	3	0	1	0	0
	4	0	0	1	0
	5	0	0	0	1

Model Convergence Status
Convergence criterion (GCONV=1E-8) satisfied.

Model Fit Statistics		
Criterion	Intercept Only	Intercept and Covariates
AIC	2610.953	1729.183
SC	2616.759	1793.044
-2 Log L	2608.953	1707.183

Initial Logistic Regression Model for Fourth Down Decisions

The LOGISTIC Procedure

Testing Global Null Hypothesis: BETA=0			
Test	Chi-Square	DF	Pr > ChiSq
Likelihood Ratio	901.7700	10	<.0001
Score	649.0058	10	<.0001
Wald	440.0012	10	<.0001

Type 3 Analysis of Effects			
Effect	DF	Wald Chi-Square	Pr > ChiSq
qtr	4	38.5977	<.0001
ydstogo	1	295.2243	<.0001
yardline_100	1	77.8875	<.0001
score_differential	1	154.8191	<.0001
game_seconds_remaini	1	8.5281	0.0035
posteam_timeouts_rem	1	4.2955	0.0382
defteam_timeouts_rem	1	0.3270	0.5674

Analysis of Maximum Likelihood Estimates						
Parameter		DF	Estimate	Standard Error	Wald Chi-Square	Pr > ChiSq
Intercept		1	4.0547	0.7664	27.9925	<.0001
qtr	2	1	-1.0464	0.2969	12.4259	0.0004
qtr	3	1	-1.4305	0.5200	7.5670	0.0059
qtr	4	1	-0.9331	0.7270	1.6472	0.1993
qtr	5	1	-1.9323	1.0846	3.1742	0.0748
ydstogo		1	-0.3713	0.0216	295.2243	<.0001
yardline_100		1	-0.0230	0.00261	77.8875	<.0001
score_differential		1	-0.0822	0.00661	154.8191	<.0001
game_seconds_remaini		1	-0.00079	0.000271	8.5281	0.0035
posteam_timeouts_rem		1	-0.1806	0.0872	4.2955	0.0382
defteam_timeouts_rem		1	0.0481	0.0841	0.3270	0.5674

Association of Predicted Probabilities and Observed Responses			
Percent Concordant	88.4	Somers' D	0.769
Percent Discordant	11.6	Gamma	0.769
Percent Tied	0.0	Tau-a	0.267
Pairs	1045845	c	0.884

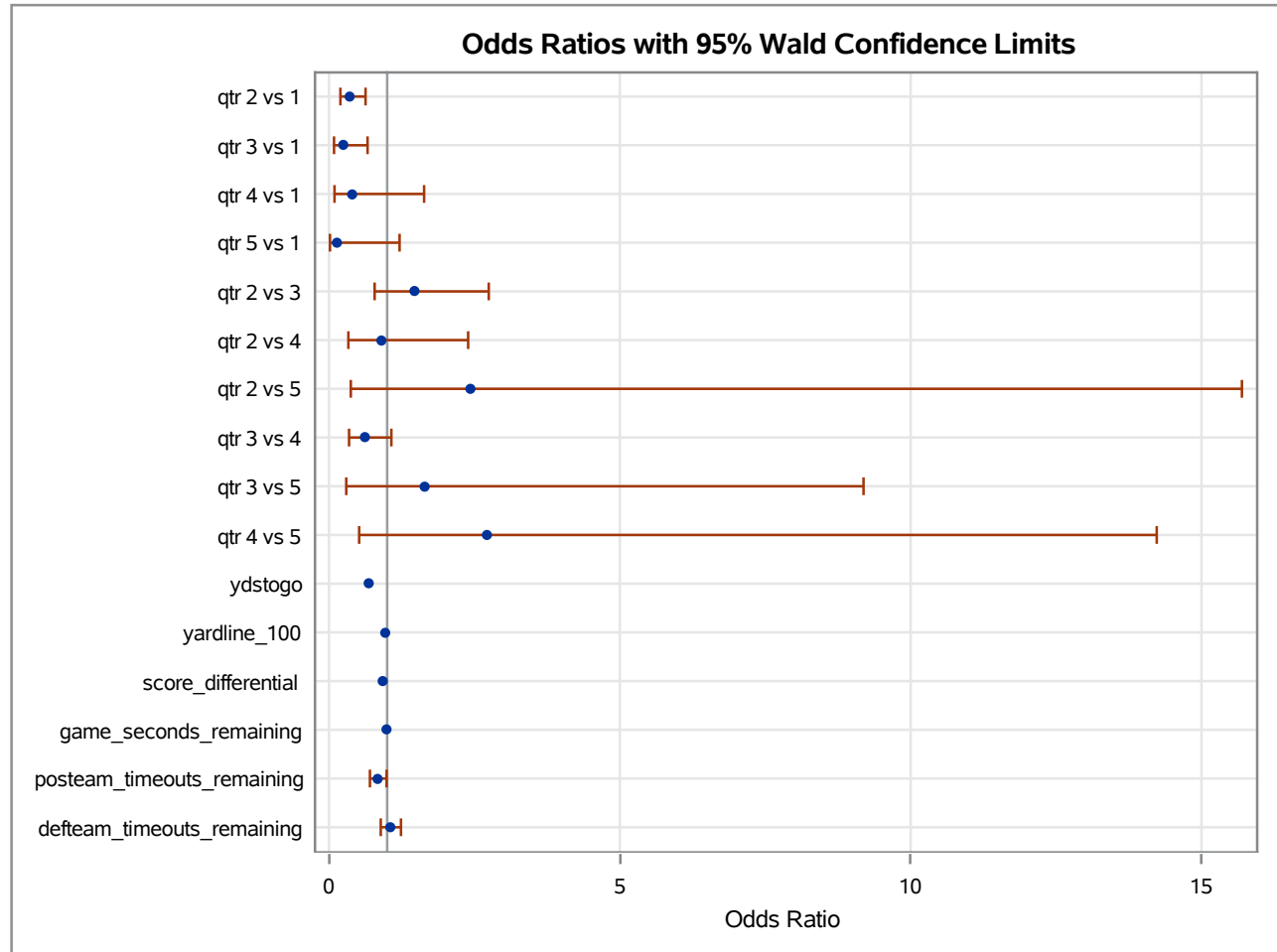
Initial Logistic Regression Model for Fourth Down Decisions

The LOGISTIC Procedure

Odds Ratio Estimates and Wald Confidence Intervals			
Odds Ratio	Estimate	95% Confidence Limits	
qtr 2 vs 1	0.351	0.196	0.628
qtr 3 vs 1	0.239	0.086	0.663
qtr 4 vs 1	0.393	0.095	1.635
qtr 5 vs 1	0.145	0.017	1.213
qtr 2 vs 3	1.468	0.785	2.746
qtr 2 vs 4	0.893	0.333	2.393
qtr 2 vs 5	2.425	0.375	15.695
qtr 3 vs 4	0.608	0.345	1.072
qtr 3 vs 5	1.652	0.297	9.189
qtr 4 vs 5	2.716	0.518	14.230
ydstogo	0.690	0.661	0.720
yardline_100	0.977	0.972	0.982
score_differential	0.921	0.909	0.933
game_seconds_remaining	0.999	0.999	1.000
posteam_timeouts_remaining	0.835	0.704	0.990
defteam_timeouts_remaining	1.049	0.890	1.237

Initial Logistic Regression Model for Fourth Down Decisions

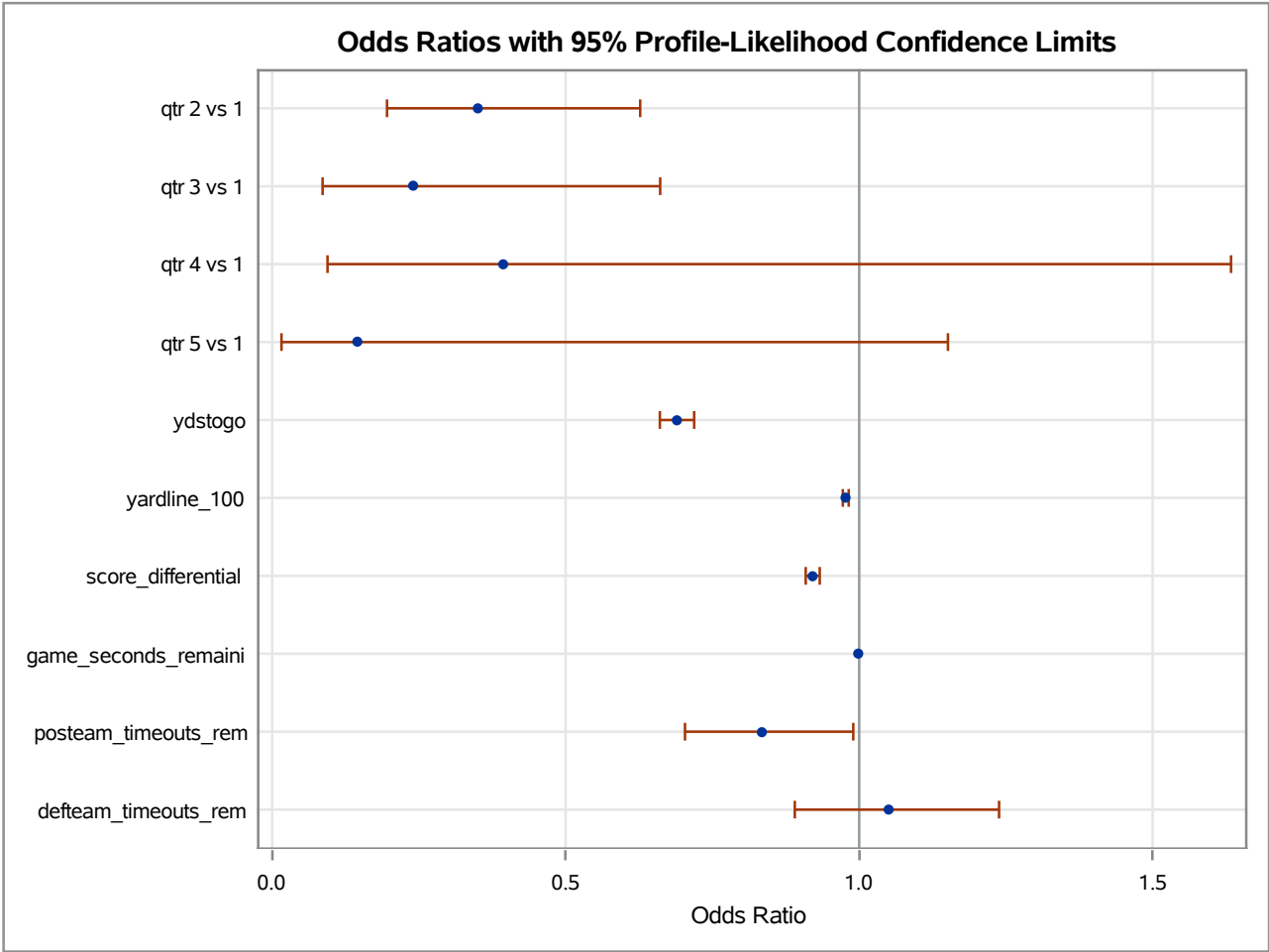
The LOGISTIC Procedure



Odds Ratio Estimates and Profile-Likelihood Confidence Intervals					
Effect		Unit	Estimate	95% Confidence Limits	
qtr	2 vs 1	1.0000	0.351	0.196	0.627
qtr	3 vs 1	1.0000	0.239	0.086	0.661
qtr	4 vs 1	1.0000	0.393	0.094	1.634
qtr	5 vs 1	1.0000	0.145	0.016	1.151
ydstogo		1.0000	0.690	0.661	0.719
yardline_100		1.0000	0.977	0.972	0.982
score_differential		1.0000	0.921	0.909	0.933
game_seconds_remaini		1.0000	0.999	0.999	1.000
posteam_timeouts_rem		1.0000	0.835	0.703	0.990
defteam_timeouts_rem		1.0000	1.049	0.890	1.238

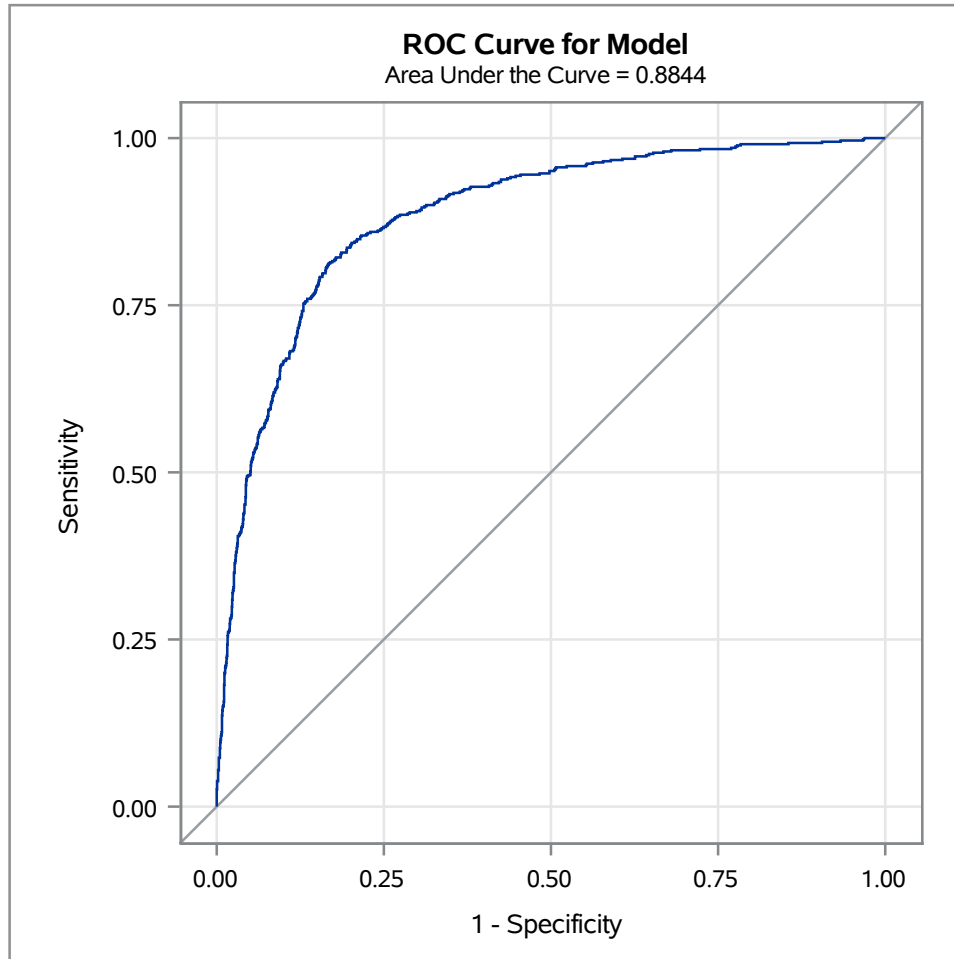
Initial Logistic Regression Model for Fourth Down Decisions

The LOGISTIC Procedure



Initial Logistic Regression Model for Fourth Down Decisions

The LOGISTIC Procedure



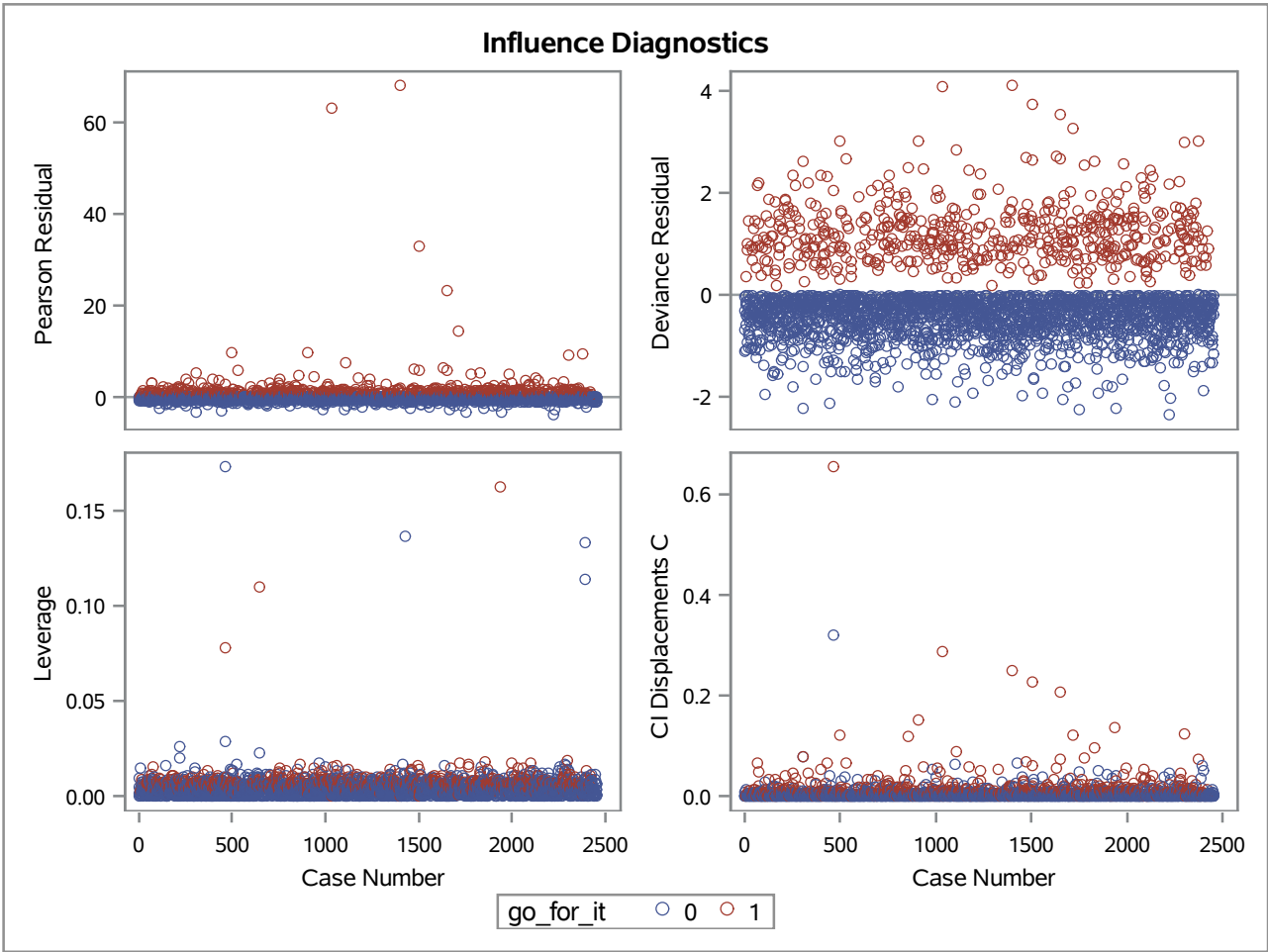
Partition for the Hosmer and Lemeshow Test					
Group	Total	go_for_it = 1		go_for_it = 0	
		Observed	Expected	Observed	Expected
1	245	4	0.28	241	244.72
2	245	5	2.04	240	242.96
3	245	8	5.83	237	239.17
4	245	10	12.09	235	232.91
5	245	14	22.71	231	222.29
6	245	26	37.26	219	207.74
7	245	45	58.73	200	186.27
8	245	102	86.99	143	158.01
9	245	137	129.35	108	115.65
10	249	198	193.73	51	55.27

Initial Logistic Regression Model for Fourth Down Decisions

The LOGISTIC Procedure

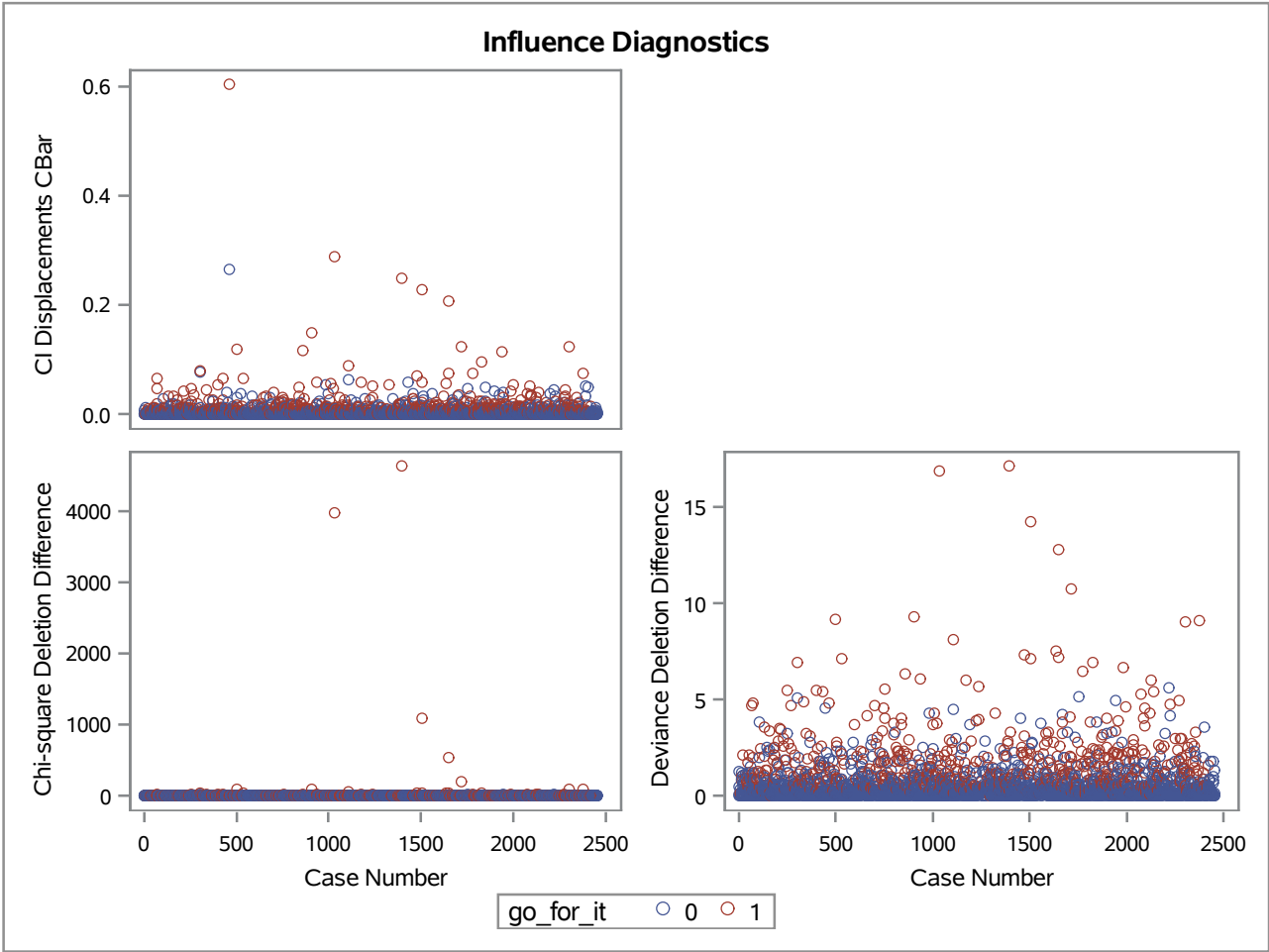
Hosmer and Lemeshow Goodness-of-Fit Test		
Chi-Square	DF	Pr > ChiSq
72.9666	8	<.0001

Classification Table									
Prob Level	Correct		Incorrect		Percentages				
	Event	Non-Event	Event	Non-Event	Correct	Sensi- tivity	Speci- ficity	Pos Pred	Neg Pred
0.500	291	1792	113	258	84.9	53.0	94.1	72.0	87.4



Initial Logistic Regression Model for Fourth Down Decisions

The LOGISTIC Procedure



Stepwise Logistic Regression Model for Fourth Down Decisions

The LOGISTIC Procedure

Model Information	
Data Set	WORK.FOURTH_DOWN_MODEL
Response Variable	go_for_it
Number of Response Levels	2
Model	binary logit
Optimization Technique	Fisher's scoring

Number of Observations Read	2454
Number of Observations Used	2454

Response Profile		
Ordered Value	go_for_it	Total Frequency
1	0	1905
2	1	549

Probability modeled is go_for_it=1.

Stepwise Selection Procedure

Class Level Information					
Class	Value	Design Variables			
qtr	1	0	0	0	0
	2	1	0	0	0
	3	0	1	0	0
	4	0	0	1	0
	5	0	0	0	1

Step 0. Intercept entered:

Model Convergence Status
Convergence criterion (GCONV=1E-8) satisfied.

-2 Log L	=	2608.953
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Residual Chi-Square Test		
Chi-Square	DF	Pr > ChiSq
649.0058	10	<.0001

Step 1. Effect ydstogo entered:

Stepwise Logistic Regression Model for Fourth Down Decisions

The LOGISTIC Procedure

Model Convergence Status
Convergence criterion (GCONV=1E-8) satisfied.

Model Fit Statistics		
Criterion	Intercept Only	Intercept and Covariates
AIC	2610.953	2100.907
SC	2616.759	2112.517
-2 Log L	2608.953	2096.907

Testing Global Null Hypothesis: BETA=0			
Test	Chi-Square	DF	Pr > ChiSq
Likelihood Ratio	512.0468	1	<.0001
Score	348.0764	1	<.0001
Wald	299.2618	1	<.0001

Residual Chi-Square Test		
Chi-Square	DF	Pr > ChiSq
367.3411	9	<.0001

Note: No effects for the model in Step 1 are removed.

Step 2. Effect score_differential entered:

Model Convergence Status
Convergence criterion (GCONV=1E-8) satisfied.

Model Fit Statistics		
Criterion	Intercept Only	Intercept and Covariates
AIC	2610.953	1941.290
SC	2616.759	1958.706
-2 Log L	2608.953	1935.290

Stepwise Logistic Regression Model for Fourth Down Decisions

The LOGISTIC Procedure

Testing Global Null Hypothesis: BETA=0			
Test	Chi-Square	DF	Pr > ChiSq
Likelihood Ratio	673.6638	2	<.0001
Score	483.2586	2	<.0001
Wald	377.3126	2	<.0001

Residual Chi-Square Test		
Chi-Square	DF	Pr > ChiSq
221.1937	8	<.0001

Note: No effects for the model in Step 2 are removed.

Step 3. Effect qtr entered:

Model Convergence Status
Convergence criterion (GCONV=1E-8) satisfied.

Model Fit Statistics		
Criterion	Intercept Only	Intercept and Covariates
AIC	2610.953	1833.514
SC	2616.759	1874.152
-2 Log L	2608.953	1819.514

Testing Global Null Hypothesis: BETA=0			
Test	Chi-Square	DF	Pr > ChiSq
Likelihood Ratio	789.4392	6	<.0001
Score	573.4061	6	<.0001
Wald	408.2627	6	<.0001

Residual Chi-Square Test		
Chi-Square	DF	Pr > ChiSq
109.9428	4	<.0001

Note: No effects for the model in Step 3 are removed.

Step 4. Effect yardline_100 entered:

Model Convergence Status
Convergence criterion (GCONV=1E-8) satisfied.

Stepwise Logistic Regression Model for Fourth Down Decisions

The LOGISTIC Procedure

Model Fit Statistics		
Criterion	Intercept Only	Intercept and Covariates
AIC	2610.953	1743.917
SC	2616.759	1790.360
-2 Log L	2608.953	1727.917

Testing Global Null Hypothesis: BETA=0			
Test	Chi-Square	DF	Pr > ChiSq
Likelihood Ratio	881.0366	7	<.0001
Score	629.9974	7	<.0001
Wald	431.3972	7	<.0001

Residual Chi-Square Test		
Chi-Square	DF	Pr > ChiSq
20.7156	3	0.0001

Note: No effects for the model in Step 4 are removed.

Step 5. Effect game_seconds_remaini entered:

Model Convergence Status
Convergence criterion (GCONV=1E-8) satisfied.

Model Fit Statistics		
Criterion	Intercept Only	Intercept and Covariates
AIC	2610.953	1729.516
SC	2616.759	1781.765
-2 Log L	2608.953	1711.516

Testing Global Null Hypothesis: BETA=0			
Test	Chi-Square	DF	Pr > ChiSq
Likelihood Ratio	897.4371	8	<.0001
Score	644.7213	8	<.0001
Wald	437.5088	8	<.0001

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Residual Chi-Square Test		
Chi-Square	DF	Pr > ChiSq
4.3318	2	0.1146

Note: No effects for the model in Step 5 are removed.

Step 6. Effect posteam_timeouts_rem entered:

Model Convergence Status
Convergence criterion (GCONV=1E-8) satisfied.

Model Fit Statistics		
Criterion	Intercept Only	Intercept and Covariates
AIC	2610.953	1727.511
SC	2616.759	1785.566
-2 Log L	2608.953	1707.511

Testing Global Null Hypothesis: BETA=0			
Test	Chi-Square	DF	Pr > ChiSq
Likelihood Ratio	901.4421	9	<.0001
Score	648.6641	9	<.0001
Wald	440.1327	9	<.0001

Residual Chi-Square Test		
Chi-Square	DF	Pr > ChiSq
0.3271	1	0.5674

Note: No effects for the model in Step 6 are removed.

Note: No (additional) effects met the 0.05 significance level for entry into the model.

Summary of Stepwise Selection							
Step	Effect		DF	Number In	Score Chi-Square	Wald Chi-Square	Pr > ChiSq
	Entered	Removed					
1	ydstogo		1	1	348.0764		<.0001
2	score_differential		1	2	151.6213		<.0001
3	qtr		4	3	118.7630		<.0001
4	yardline_100		1	4	90.1077		<.0001

Stepwise Logistic Regression Model for Fourth Down Decisions

The LOGISTIC Procedure

Summary of Stepwise Selection							
Step	Effect		DF	Number In	Score Chi-Square	Wald Chi-Square	Pr > ChiSq
	Entered	Removed					
5	game_seconds_remaini		1	5	16.2846		<.0001
6	posteam_timeouts_rem		1	6	4.0103		0.0452

Type 3 Analysis of Effects			
Effect	DF	Wald Chi-Square	Pr > ChiSq
qtr	4	38.3004	<.0001
ydstogo	1	295.4533	<.0001
yardline_100	1	77.7187	<.0001
score_differential	1	154.4739	<.0001
game_seconds_remaini	1	8.3480	0.0039
posteam_timeouts_rem	1	3.9978	0.0456

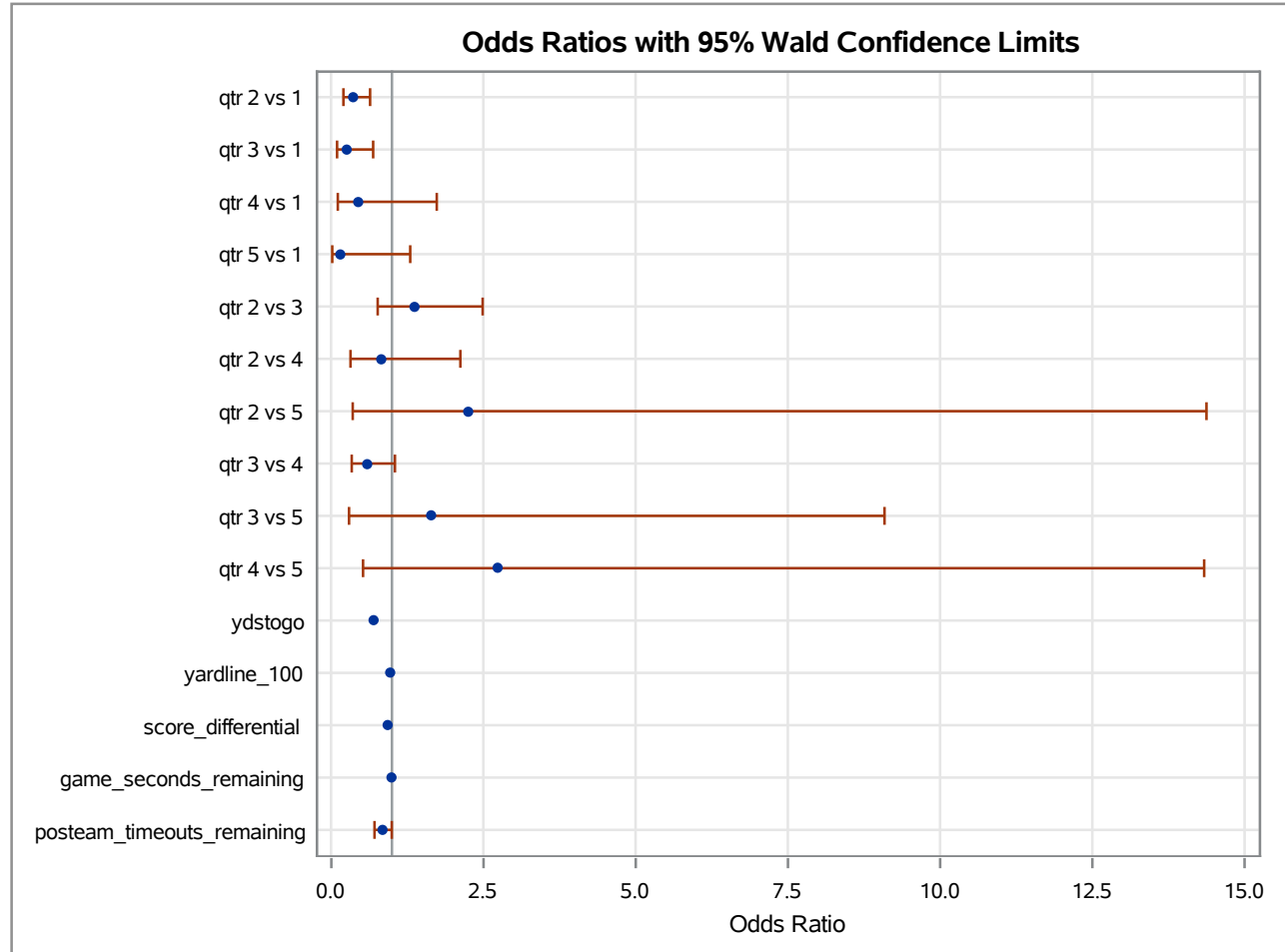
Analysis of Maximum Likelihood Estimates						
Parameter		DF	Estimate	Standard Error	Wald Chi-Square	Pr > ChiSq
Intercept		1	4.0077	0.7618	27.6759	<.0001
qtr	2	1	-1.0190	0.2930	12.0923	0.0005
qtr	3	1	-1.3413	0.4961	7.3100	0.0069
qtr	4	1	-0.8252	0.7024	1.3803	0.2401
qtr	5	1	-1.8348	1.0700	2.9403	0.0864
ydstogo		1	-0.3714	0.0216	295.4533	<.0001
yardline_100		1	-0.0230	0.00261	77.7187	<.0001
score_differential		1	-0.0821	0.00661	154.4739	<.0001
game_seconds_remaini		1	-0.00074	0.000257	8.3480	0.0039
posteam_timeouts_rem		1	-0.1703	0.0852	3.9978	0.0456

Association of Predicted Probabilities and Observed Responses			
Percent Concordant	88.4	Somers' D	0.769
Percent Discordant	11.6	Gamma	0.769
Percent Tied	0.0	Tau-a	0.267
Pairs	1045845	c	0.884

Stepwise Logistic Regression Model for Fourth Down Decisions

The LOGISTIC Procedure

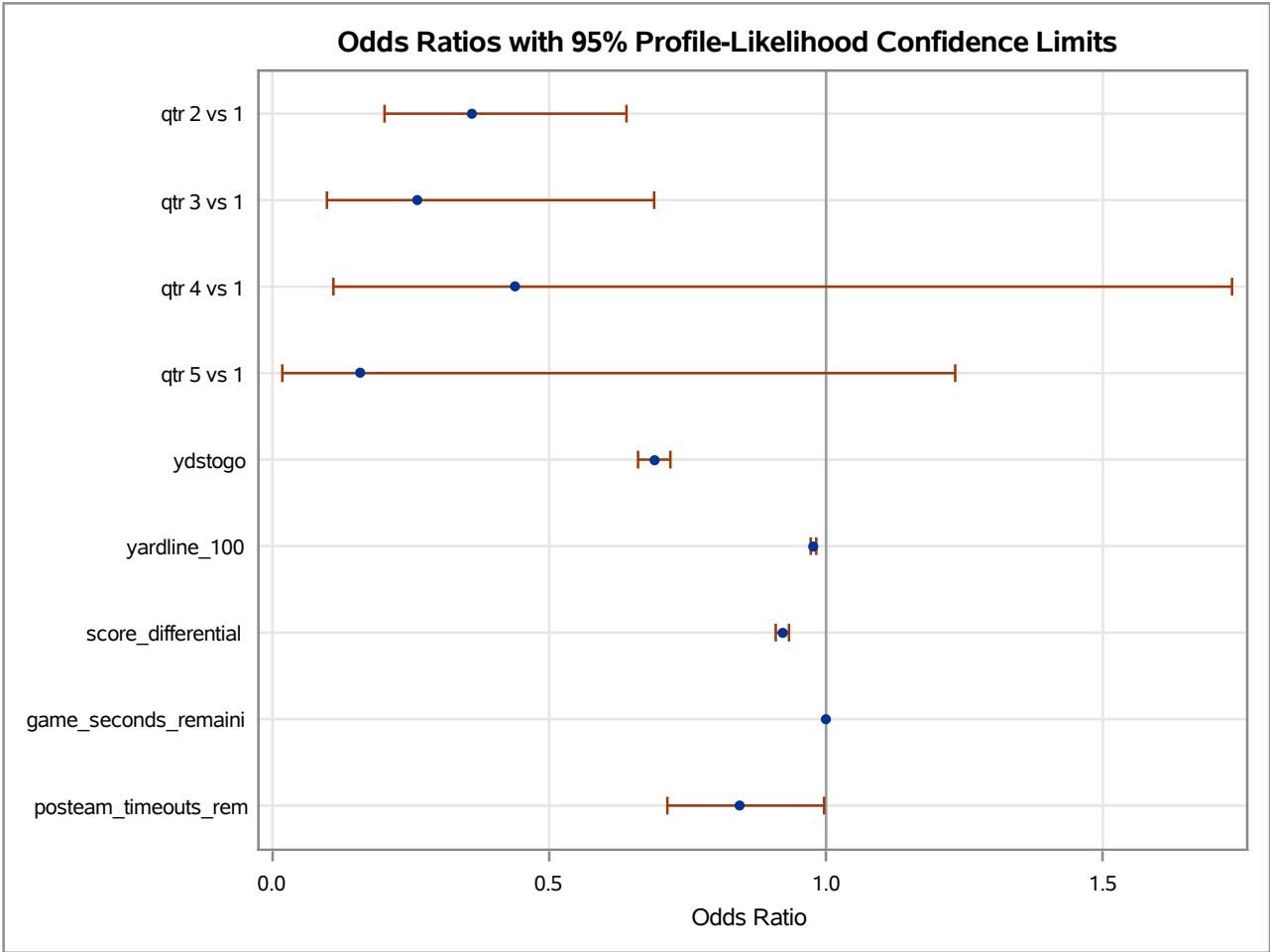
Odds Ratio Estimates and Wald Confidence Intervals			
Odds Ratio	Estimate	95% Confidence Limits	
qtr 2 vs 1	0.361	0.203	0.641
qtr 3 vs 1	0.261	0.099	0.691
qtr 4 vs 1	0.438	0.111	1.736
qtr 5 vs 1	0.160	0.020	1.300
qtr 2 vs 3	1.380	0.766	2.489
qtr 2 vs 4	0.824	0.320	2.123
qtr 2 vs 5	2.261	0.356	14.374
qtr 3 vs 4	0.597	0.340	1.049
qtr 3 vs 5	1.638	0.295	9.088
qtr 4 vs 5	2.744	0.525	14.336
ydstogo	0.690	0.661	0.720
yardline_100	0.977	0.972	0.982
score_differential	0.921	0.909	0.933
game_seconds_remaining	0.999	0.999	1.000
posteam_timeouts_remaining	0.843	0.714	0.997

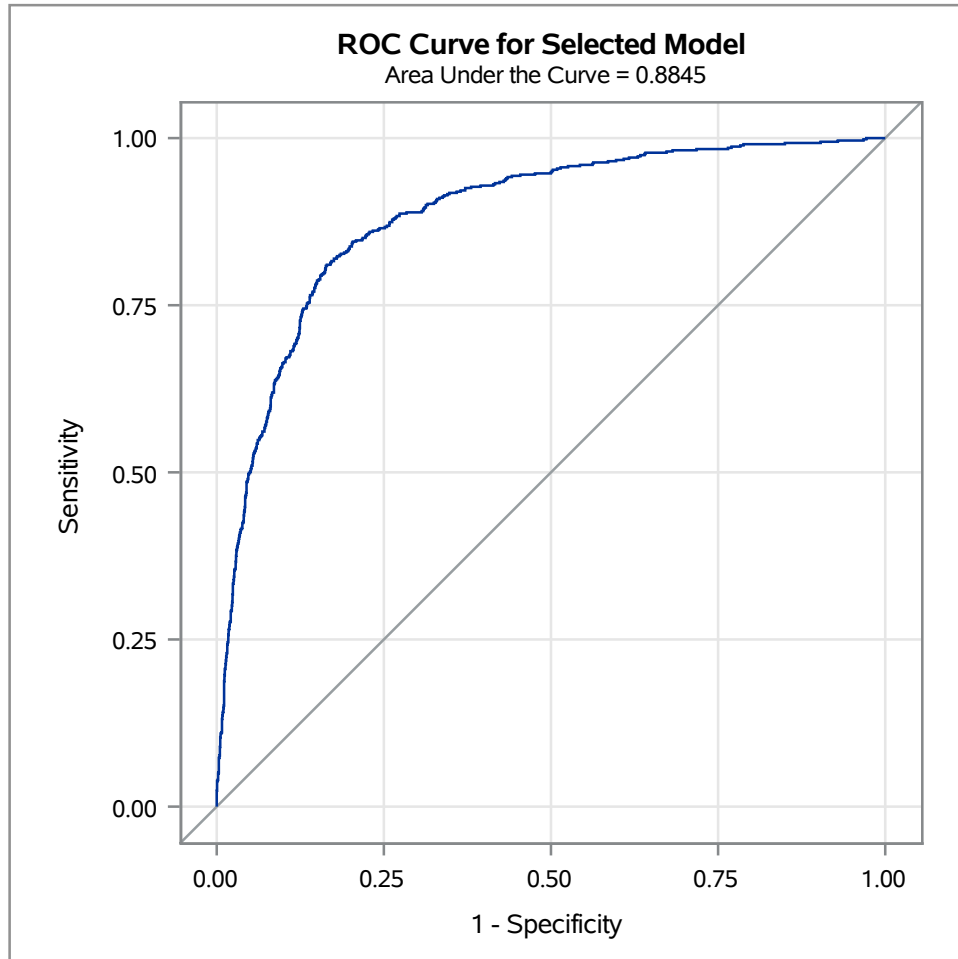


Stepwise Logistic Regression Model for Fourth Down Decisions

The LOGISTIC Procedure

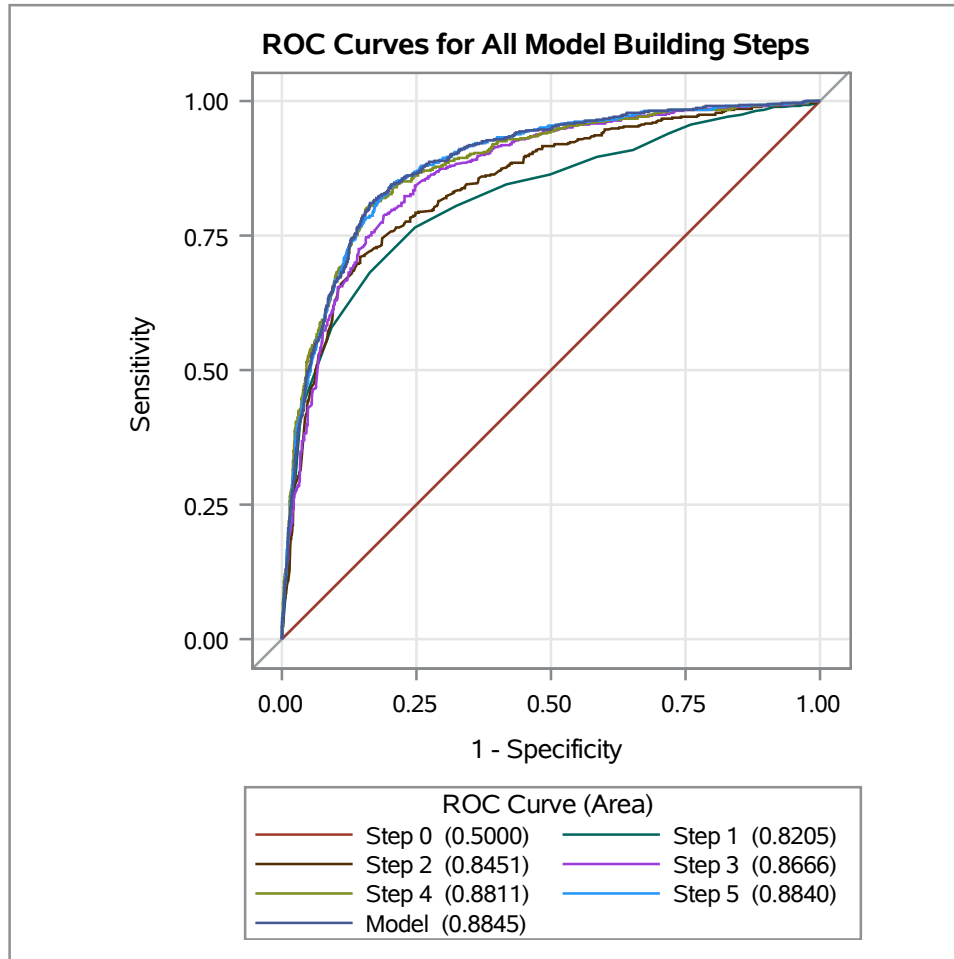
Odds Ratio Estimates and Profile-Likelihood Confidence Intervals					
Effect		Unit	Estimate	95% Confidence Limits	
qtr	2 vs 1	1.0000	0.361	0.203	0.640
qtr	3 vs 1	1.0000	0.261	0.099	0.690
qtr	4 vs 1	1.0000	0.438	0.110	1.733
qtr	5 vs 1	1.0000	0.160	0.018	1.233
ydstogo		1.0000	0.690	0.660	0.719
yardline_100		1.0000	0.977	0.972	0.982
score_differential		1.0000	0.921	0.909	0.933
game_seconds_remaini		1.0000	0.999	0.999	1.000
posteam_timeouts_rem		1.0000	0.843	0.713	0.996



Stepwise Logistic Regression Model for Fourth Down Decisions**The LOGISTIC Procedure**

Stepwise Logistic Regression Model for Fourth Down Decisions

The LOGISTIC Procedure



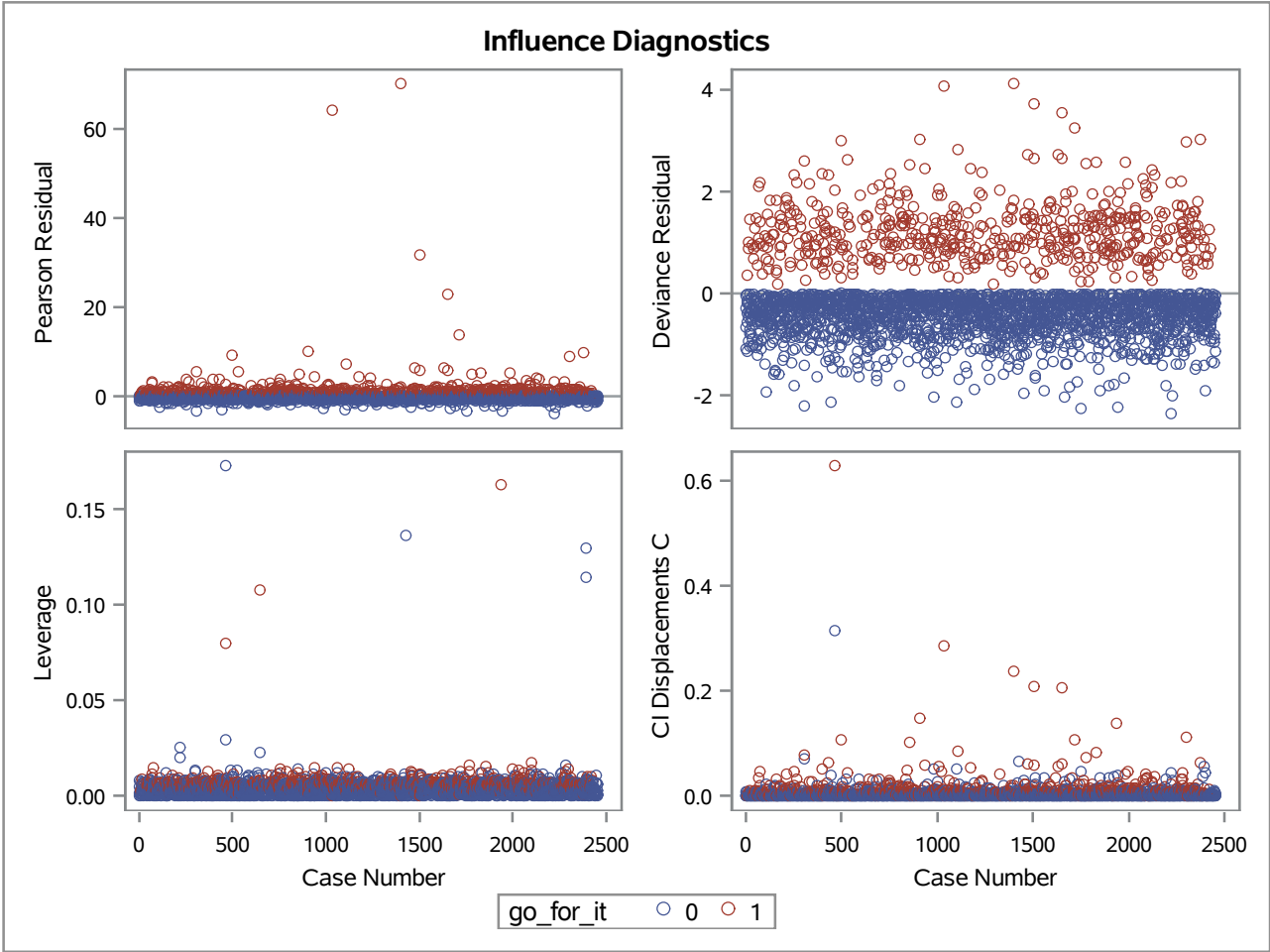
Partition for the Hosmer and Lemeshow Test					
Group	Total	go_for_it = 1		go_for_it = 0	
		Observed	Expected	Observed	Expected
1	245	4	0.28	241	244.72
2	245	5	2.04	240	242.96
3	245	7	5.84	238	239.16
4	245	12	12.06	233	232.94
5	245	13	22.73	232	222.27
6	245	26	37.22	219	207.78
7	245	45	58.85	200	186.15
8	245	99	86.90	146	158.10
9	245	143	129.41	102	115.59
10	249	195	193.68	54	55.32

Stepwise Logistic Regression Model for Fourth Down Decisions

The LOGISTIC Procedure

Hosmer and Lemeshow Goodness-of-Fit Test		
Chi-Square	DF	Pr > ChiSq
73.4462	8	<.0001

Classification Table									
Prob Level	Correct		Incorrect		Percentages				
	Event	Non-Event	Event	Non-Event	Correct	Sensi- tivity	Speci- ficity	Pos Pred	Neg Pred
0.500	292	1793	112	257	85.0	53.2	94.1	72.3	87.5



Stepwise Logistic Regression Model for Fourth Down Decisions

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